

Digital Logic Design (EE1005)

Date: 27th May, 2025

Course Instructor(s)

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Final Exam

Total Time (Hrs): 3

Total Marks: 120

Total Questions: 6

Roll No

Section

Student Signature

- Attempt all the questions.
- Show complete working of each question.
- Multiple solutions of the same question will carry zero credit.
- State your valid assumptions clearly if you have to take any.

CLO #2: Apply Boolean Algebra and K-map methods to optimize logic circuits

Q1: [20 marks]

- a. Use K-Map method to minimize the following function in Sum of Product form

$$F(W,X,Y,Z) = \sum m(0, 2, 4, 5, 8, 14, 15), d(W,X,Y,Z) = \sum m(7, 10, 13)$$

- b. Implement two Input XNOR gate using two input NOR Gates only and apply gate reduction to reduce the circuit.

CLO #3: Design combinational circuits using functional blocks

Q2: [20 marks]

Design a combinational circuit with two data inputs A and B (each of two bits) i.e. A= A1 A0 and B= B1 B0 and two selection inputs, S1, S0. The system should implement the following functions based on selection inputs

S1	S0	Operation
0	0	F=0
0	1	$F = \begin{cases} 1, & \text{if } (A + B) < 2 \\ 0, & \text{Otherwise} \end{cases}$
1	0	$F = \begin{cases} 1, & \text{if total number of 1s in both A and B are odd} \\ 0, & \text{Otherwise} \end{cases}$
1	1	F=1

- Construct the truth table of the functions and use decoders of appropriate size (if required) to implement the functions.
- Then use a Multiplexer to generate the output. Construct a complete circuit diagram.

Use of at least two Decoders and one Multiplexer is mandatory in your design

Available resources are:

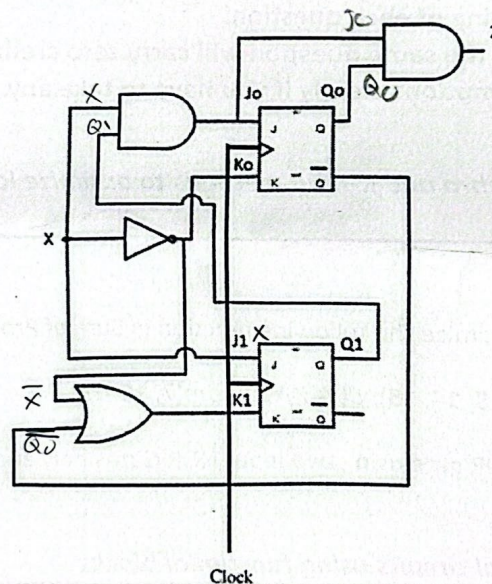
- Decoders and Multiplexers of all sizes
- Two, three and four input AND, OR, NAND and NOR gates
- 2 input XOR and XNOR gates
- NOT gates

CLO #4: Analyze elements of sequential circuits

Q3:

[20 marks]

Analyze the logic circuit given below



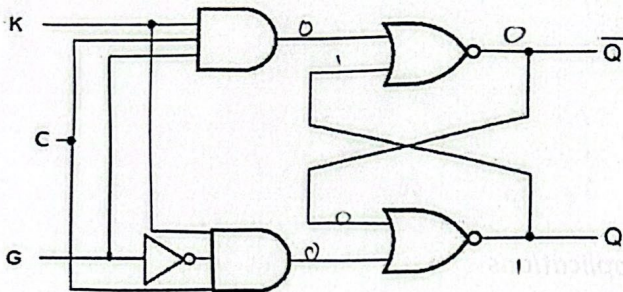
- Find the Boolean expression for J_0 , K_0 , J_1 , and K_1
- Construct the state table and state diagram.
- State if it is a Mealy or Moore Model circuit.

CLO #4: Analyze elements of sequential circuits

Q4:

[20 marks]

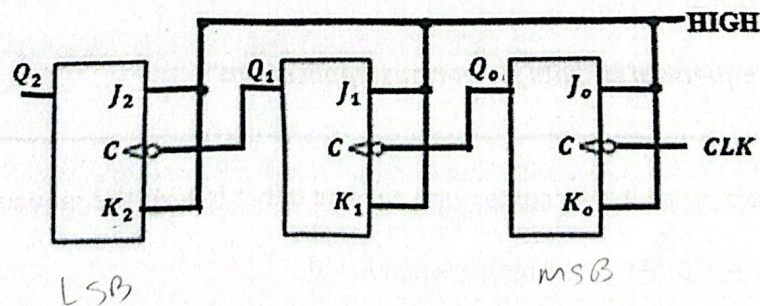
- a. Circuit diagram of a new type of latch called KG latch is shown in the figure below.

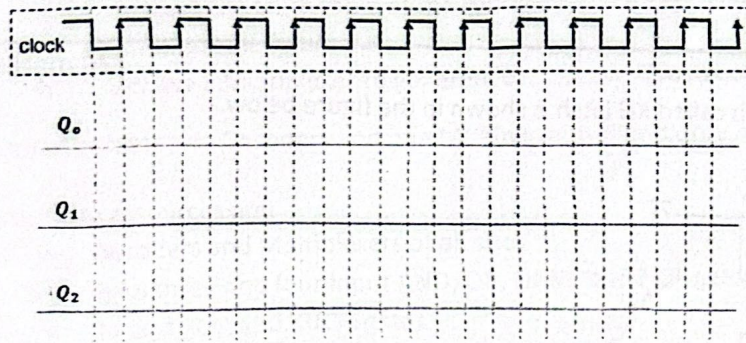


Analyze the circuit and fill-in the Next State Column of the function table of KG Latch. Identify if it is set, reset, hold or undefined (if applicable) state. Derive the expression $Q(t+1)$ as a function of $Q(t)$, K , and G .

Latch Inputs		Next State
K	G	$Q(t+1)$
0	0	
0	1	
1	0	
1	1	

- b. **Analyze** the circuit to generate the timing diagram of the circuit. Assume all flip-flops have zero initial value. Show the contents of Q_0 , Q_1 and Q_2 for at least 6 clock cycles in the given timing diagram.



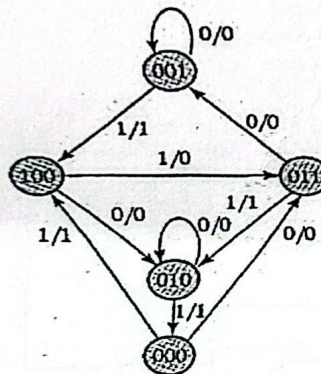


CLO #5: Design of sequential circuits for various applications

Q5:

[20 marks]

Design a sequential circuit using D Flip flop that implements the following state diagram.



- Construct the state table. Take unused states as don't care.
- Find the expressions for flip-flop inputs and construct the logic circuit.
- Construct the state diagram corresponding to your circuit including the unused states.

CLO #5: Design of sequential circuits for various applications

Q6:

[20 marks]

Design a synchronous binary arbitrary counter with an input R that follows the sequence given below

- $2 \rightarrow 4 \rightarrow 6 \rightarrow 0 \rightarrow 7$ and repeat when $R = 0$
- $0 \rightarrow 2 \rightarrow 6 \rightarrow 4 \rightarrow 7$ and repeat when $R = 1$

- Construct the state diagram and state table.
- Construct the logic diagram of your design using T- flip flops and logic gates of your choice.